

Espressif Documentation MCP Server: Power Your AI Agents with Espressif Docs

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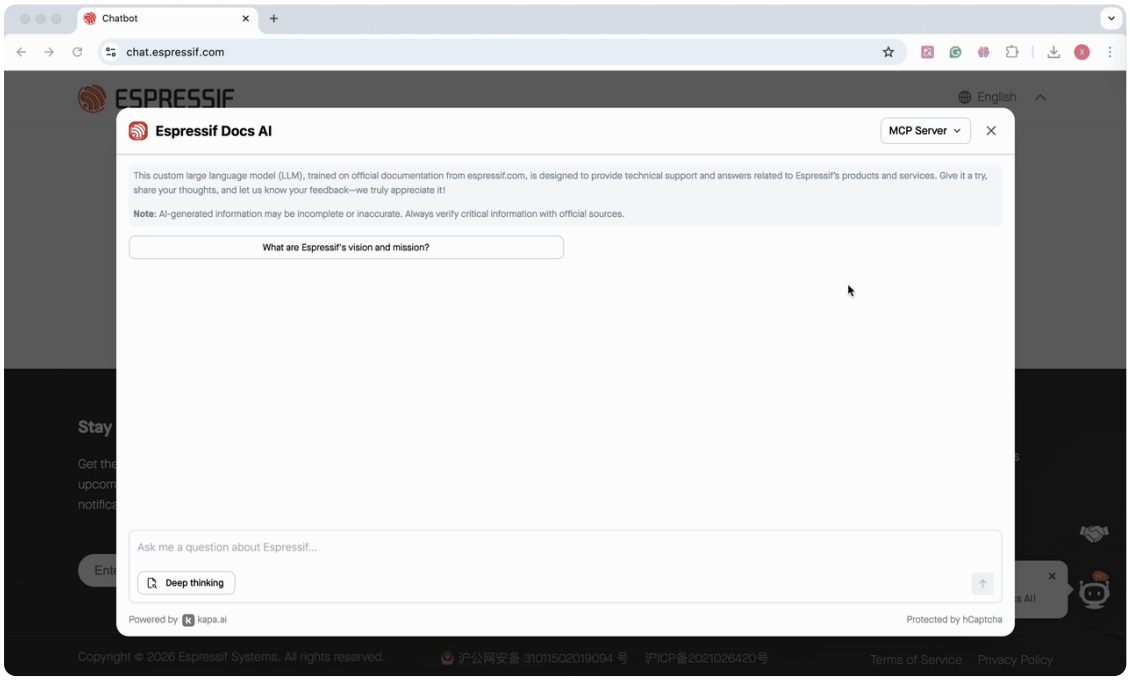
> ***This article introduces how to connect your AI agent to official, up-to-date Espressif documentation directly inside your AI applications — with installation steps, example prompts, and best practices.***

Why MCP?

Model Context Protocol (MCP) is a standard that lets AI agents connect to external data sources, tools, and workflows. Instead of guessing from training data or search results, AI agents act on real, up-to-date context.

The Espressif Documentation MCP server implements this standard for Espressif's documentation. Once installed in Cursor, VS Code, Claude Code, or

- **Stay in your AI applications** — query the full Espressif documentation corpus through your AI agents without switching to a browser or a PDF viewer
- **Work faster and more accurately** — get implementation guidance, code examples, and bugfix suggestions grounded in official documentation, rather than relying on web search results or agent training data
- **Reduce hallucinations** — agent outputs are based on real, current Espressif documentation rather than fabricated or obsolete content



MCP Server Capabilities

The Espressif Documentation MCP server provides the following tools to AI agents:

Tool	Description
`search_espressif_sources (query, language)`	Performs semantic retrieval against the latest Espressif documentation in English and Chinese. It

URLs, providing official reference for your AI assistants.

The MCP server shares the same knowledge sources as the Espressif Documentation chatbot, including:

- Chip and module datasheets
- Technical Reference Manuals (TRM)
- Hardware Design Guidelines (HDG)
- SDK documentation, such as the latest version of ESP-IDF Programming Guide
- Product Change Notices (PCN), advisories, and certificates
- Espressif blog posts and technical articles
- Documentation for selected end-of-life (EOL) or not-recommended-for-new-design (NRND) products
- Selected M5Stack documentation

MCP Server vs. Chatbot

Both the MCP server and the Espressif Documentation chatbot provide access to up-to-date Espressif documentation, but they are designed for different scenarios.

Use Case	MCP Server	Documentation Chatbot
Integrated into IDE workflow	✅	❌
Modify project code	✅	❌
Perform structured reasoning over docs	✅	⚠️ Limited
Quick documentation Q&A	⚠️	✅

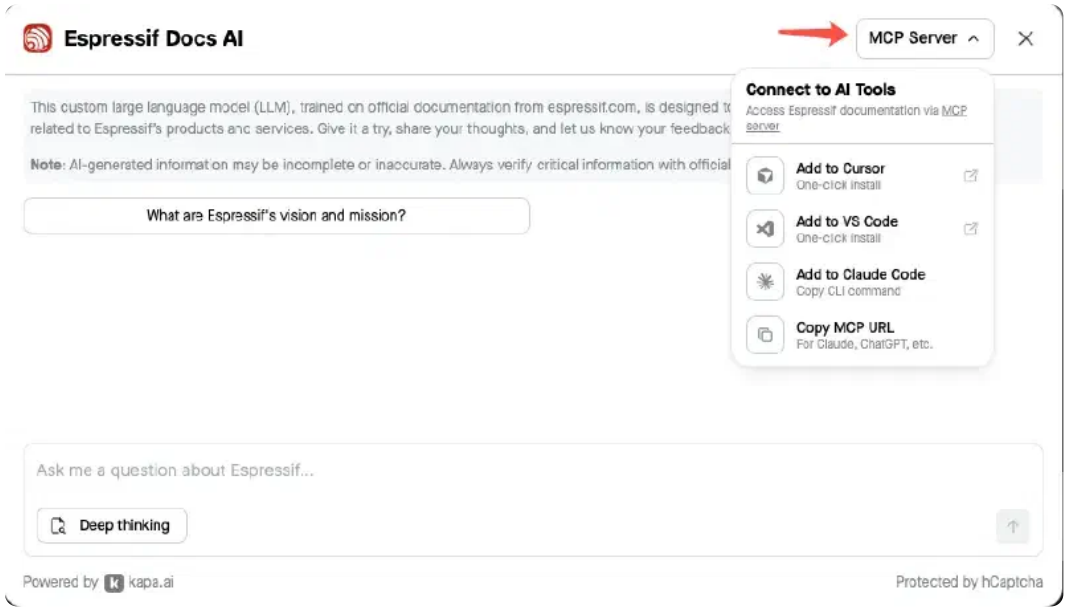
Non-technical users	❌	✅

In short:

- Use the **MCP server** for integrating documentation into your development workflow.
- Use the **chatbot** for learning and documentation exploration.

How to Install the MCP Server

1. Open the Espressif Documentation chatbot widget
2. Click "**MCP Server**" in the upper-right of the widget. Choose where to use the MCP server and follow the corresponding instructions below.



Cursor | VS Code | Claude Code | Claude Desktop | Antigravity | Antigravity CLI | Gemini CLI | Codex | OpenCode | Other

2. In the Espressif Documentation chatbot widget under **"MCP Server"**, click **"Add to Cursor"**. You may need to allow the website to open Cursor.

▼ If clicking "Add to Cursor" does nothing:

Add the following to your `~/.cursor/mcp.json` file:

```
{
  "mcpServers": {
    "espressif-docs": {
      "url": "https://mcp.espressif.com/docs"
    }
  }
}
```

Save the file, restart Cursor, and open the Cursor Settings page. Continue from Step 4.

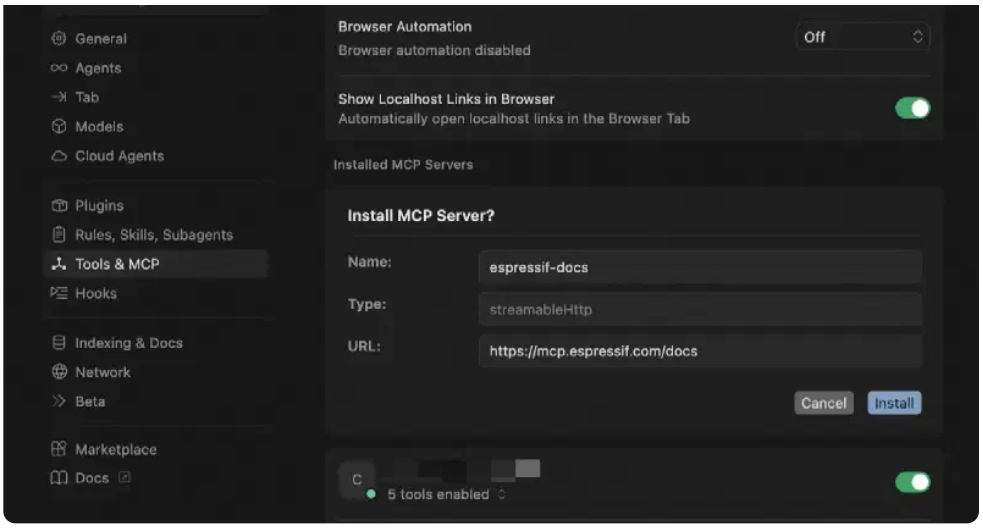
On Linux, clicking **Add to Cursor** or editing the `~/.cursor/mcp.json` file may have no effect if the system has not registered the `cursor://` URL protocol handler.

To verify, run the following command in a terminal:

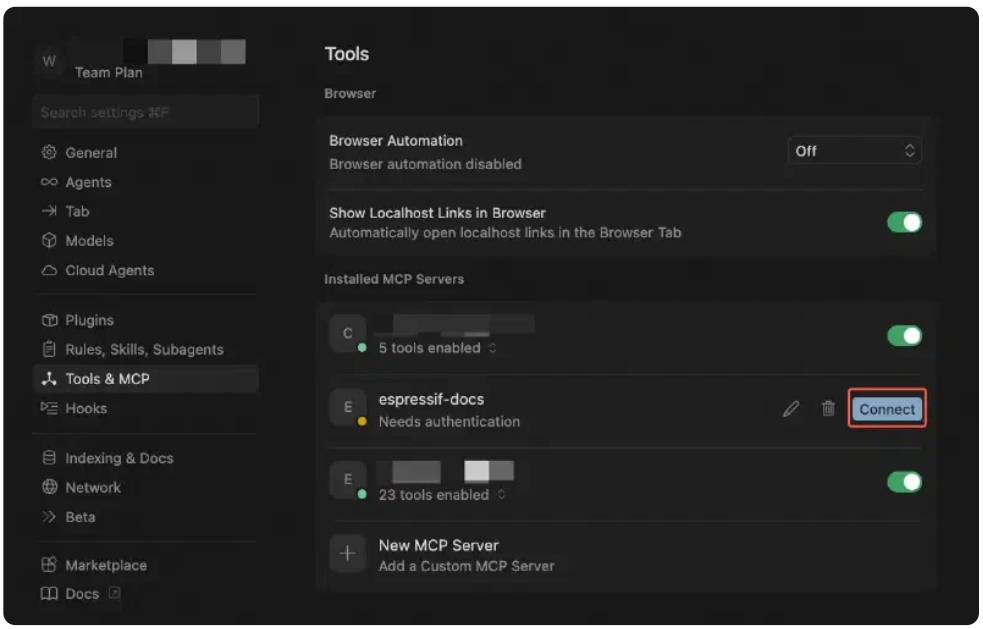
```
xdg-open "cursor://test"
```

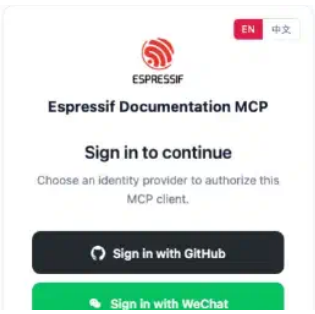
If nothing happens or an error is shown, the URL handler is not registered. This is a system-level issue and must be fixed before the MCP server can be added via browser.

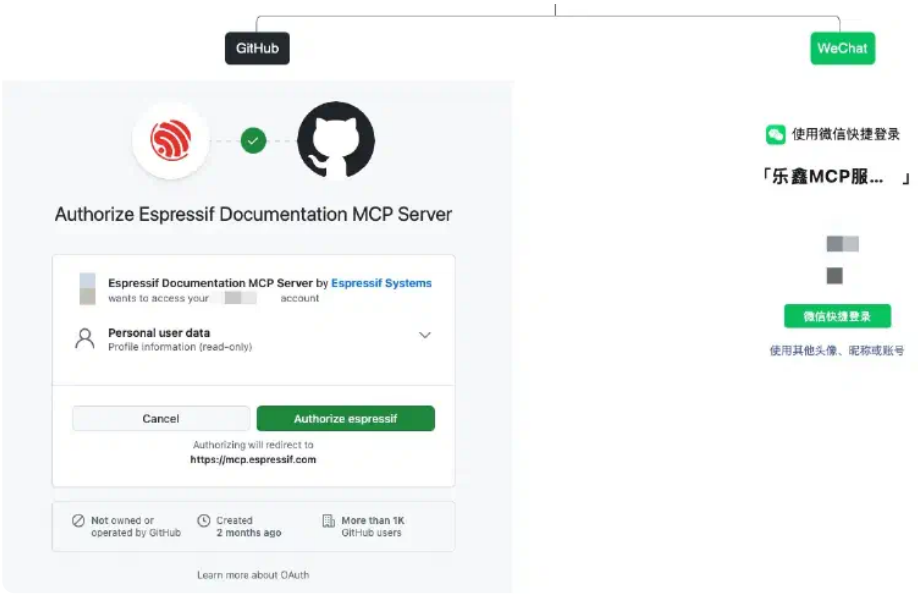
3. In the Cursor Settings page that opens, click **"Install"** under **"Install MCP Server?"**



4. After the MCP server is installed, click "**Connect**" to authenticate. Allow Cursor to open the external website, and sign in with GitHub or WeChat to authenticate with the MCP server.







Example Prompts

You can start experimenting with the Documentation MCP server for the following use cases:

Generate code

> *I have an ESP32-S3-Box-3 and an external LED. I want to write a C application using ESP-IDF to control the LED intensity via LEDC (PWM). Please provide a detailed guide and a code example, including the GPIO configuration.*

Review code

> *Review the SPI initialization code against the ESP-IDF SPI Master driver documentation. Highlight and correct any wrong API usage, deprecated functions, or missing configuration steps.*

Troubleshoot

> *this in the ESP-IDF documentation and suggest a fix.*

Update code following the style guide

> *Update the code to follow the Espressif IoT Development Framework Style Guide.*

Generate files

> *Based on the ESP-IDF partition table documentation, generate a partition table CSV for a 4 MB flash with a 1 MB factory app partition, a 512 KB OTA_0 partition, a 512 KB OTA_1 partition, and a 16 KB NVS partition.*

Update configuration options

> *Check Espressif documentation for the recommended I2C pins on ESP32-C3, and modify pin configuration options* `I2C_MASTER_SCL` *and* `I2C_MASTER_SDA` *accordingly.*

Migrate to a new ESP-IDF version

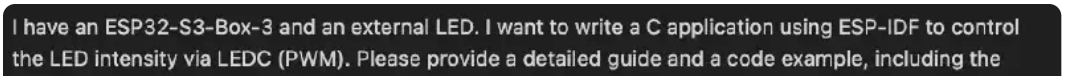
> *My project currently targets ESP-IDF v5.1. Using the ESP-IDF v5.2 and v5.3 migration guides, list the breaking API changes that affect UART and I2C drivers, and update the calls in* `main/comm.c` *accordingly.*

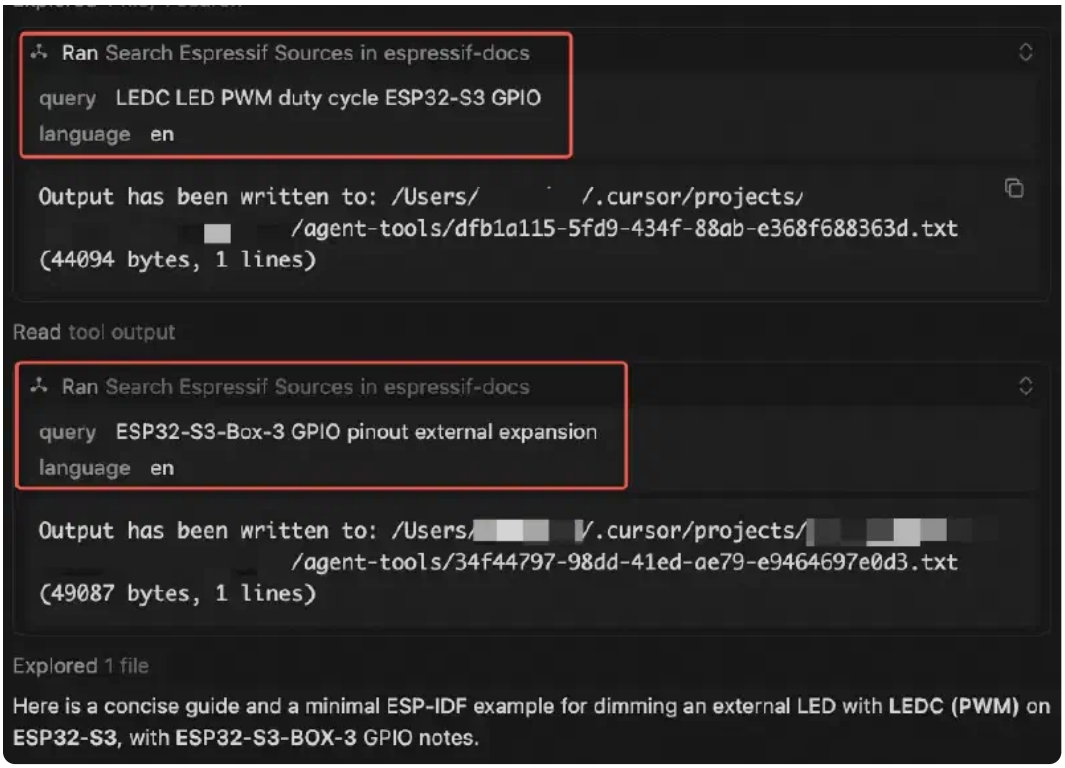
Best Practices

- **Check the tool call indicator** to confirm the MCP server is being queried. If you don't see it, the agent may be answering from training data or searching the web.









- **Explicitly instruct the agent** if it does not consult the MCP server by default. Add `"refer to Espressif documentation"` to your prompt, or define this requirement in `AGENTS.md` :

```
# AGENTS.md
Always use the Espressif documentation MCP server if you need to work
with ESP chips or ESP SDKs such as ESP-IDF and ESP-ADF, without me
having to explicitly ask.
```

Limitations

The Espressif Documentation MCP server has the following limitations:

- **Retrieval only** — the MCP server retrieves documentation and supplies it as context for AI agents. It does not execute code, modify files, or perform actions — those remain the responsibility of the AI agent.

repositories, GitHub issues, community or third-party forums, internal or unpublished documents, or documentation for all EOL and NRND products.

- **Rate limits apply** — to ensure fair access for all users, the following per-user limits apply:
  - 40 requests per user per hour
  - 200 requests per user per day
- **Not designed for open-ended learning or exploration** — if you want to ask questions, browse topics, or get documentation summaries conversationally, use the Espressif Documentation Chatbot instead.

FAQ

Do I need internet access to use the MCP server?

Yes. You need access to the public internet to connect to the MCP server.

Do I need an account to use the MCP server?

Yes. You need either a GitHub account or a WeChat account to authenticate with the MCP server. Only your anonymized account ID is stored — this is used solely to enforce per-user rate limits and ensure fair usage.

How many documentation MCP servers does Espressif have? Do I need to add them separately for each entry point?

Espressif has only one documentation MCP server at `https://mcp.espressif.com/docs`. The entry points below are simply different ways to add it.

- The AI chatbot widget on chat.espressif.com

- The Espressif MCP Servers page

The default MCP server name may differ slightly depending on which entry point you use, but they all connect to the same underlying service.

When should I use the MCP server?

AI agents call the MCP server automatically when they need context they don't already have. If the answer can be found in your open project files, the agents will use those instead.

For example, if you have ESP-IDF open in your application, asking about the ESP-IDF programming guide will lead the agent to consult the local source code and `docs/` folder, not the MCP server.

Do I need to manually start the MCP server after installation?

No. The MCP server is a remote service, so there is no local process to start.

Exception: If you configured the MCP server by manually editing `mcp.json` in VS Code, click **Start** above the server entry to open the authentication page, see VS Code. Once authenticated, the server connects automatically from that point on.

Why didn't the agent consult the Espressif Documentation MCP server, even though I asked it to?

Step 1: Check the MCP server status in your application.

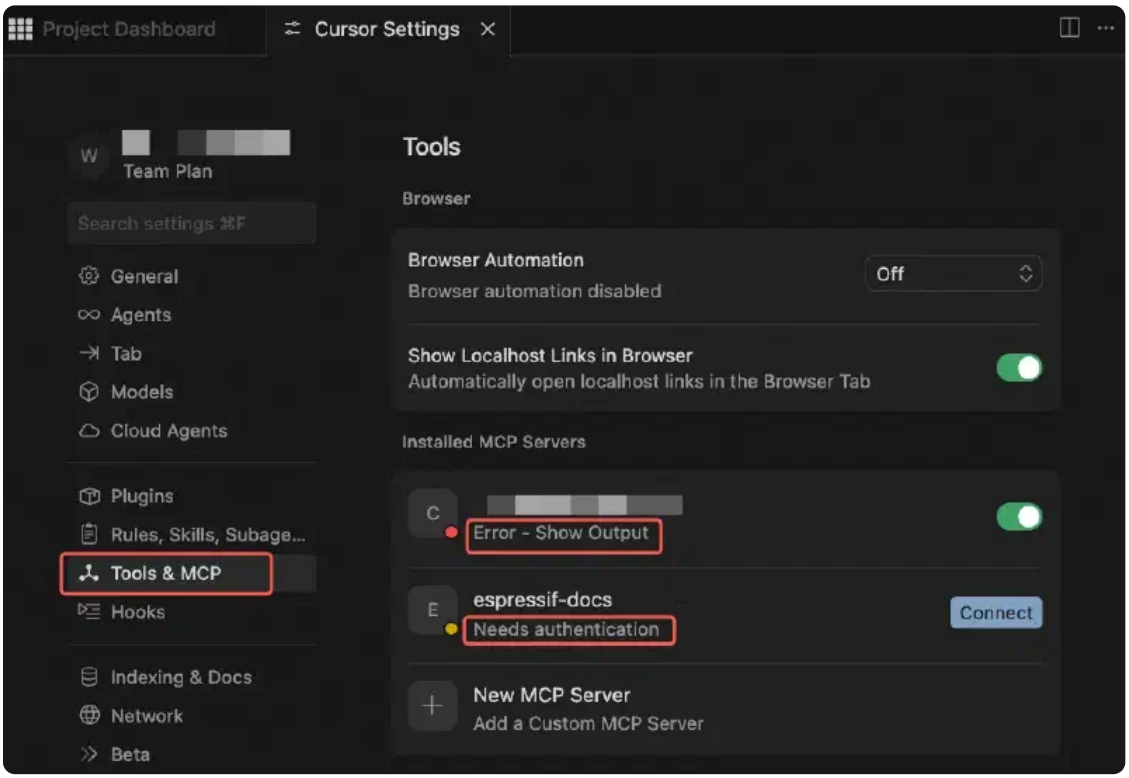
- **Cursor**: Go to **Settings** > **Tools & MCP**.
- **VS Code**: Open the Extensions panel and go to **MCP SERVERS – INSTALLED**.
- **Claude Code**: Run `claude mcp list` command.

- **Antigravity**: Go to **Antigravity Settings** > **Customizations** > **INSTALLED MCP SERVERS**.
- **Antigravity CLI**: Run `agy` to launch Antigravity CLI, and type `/mcp`.
- **Gemini CLI**: Run `gemini mcp list` command.
- **Codex**: Run `codex mcp list` command.
- **OpenCode**: Run `opencode mcp list` command.

If the MCP server is disabled, enable it.

If authentication has expired, re-authenticate or re-add the MCP server.

If an error persists, submit a feedback with the error message shown in your AI application.



Step 2: Check whether the agent is allowed to use the MCP server tools.

- **Cursor**: Go to **Settings** > **Agents** > **MCP Allowlist**. The agent will prompt

- **VS Code**: Agent tool permissions are managed per-session. The first time the agent calls an MCP tool in a session, it will prompt you to allow or deny it.
- **Claude Code**: The first time the agent calls an MCP tool, it will prompt you to allow or deny it. You may also run:

```
claude --allowedTools mcp__espressif-docs__search_espressif_sources
```

- **Claude Desktop**: Go to **Settings** > **Customize** > **Connectors** > **Desktop** > **Tool permissions**. By default, this is set to `Needs approval`, meaning you will be prompted each time the agent calls a tool.
- **Antigravity**: Enabled by default. Go to **Antigravity Settings** > **Customizations** > **INSTALLED MCP SERVERS**, and check "**tool enabled**".
- **Antigravity CLI**: The first time the agent calls an MCP tool, it will prompt you to allow or deny it.
- **Gemini CLI**: Enabled by default.
- **Codex**: Enabled by default.
- **OpenCode**: Enabled by default.

Step 3. Try a different AI application.

MCP support varies across AI application. If the MCP server is enabled and allowed but the agent still does not call it, try a different application. For example, if you are using Cursor, try VS Code or Claude Code instead.

If the issue persists across application, the problem likely lies with the MCP server itself. Check the status page to see if there are any known outages.

Can the MCP server answer questions about end-of-life software, e.g., ESP-IDF v4.4?

latest, is not included. If your project targets an EOL version, the MCP server may return documentation that does not match your SDK version.

Why is the agent linking to documents in English instead of Chinese?

The MCP server searches either the English or Chinese documentation depending on the language of your query. If your AI agents do not specify the query language, English documentation is returned by default.

Each response includes links to the source documentation pages, so you can refer to the full context of the returned content.

Related Resources

- 📺 Video: Espressif Documentation MCP Server Getting Started — A complete walkthrough covering installation, GitHub authentication, and migrating a project from ESP-IDF 5.1 to 6.0 using MCP-retrieved documentation.